The Scientific Event Horizon: Bounded-Error Simulation via Specification-Driven Agent Pipelines across Scientific Domains

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Summary. Scientists in every discipline build bespoke computational solvers—finite-element codes for bridges, quantum-chemistry packages for molecules, epidemiological simulators for disease spread. Each solver works for one class of problems but cannot be reused for another, creating an expertise bottleneck that limits who can simulate what. Here we ask: is there a single framework that can simulate *any* problem within a rigorously defined class, with guaranteed error bounds? We identify such a class—the *simulability class* $\mathfrak{S}$, comprising all finitely specifiable, numerically approximable problems with explicit observables—and prove its sharp boundary against four fundamental mathematical obstructions. We show that structured specifications (`spec.md` files), validated and executed by a three-agent pipeline over 12 computational primitives, achieve bounded-error simulation for every problem in $\mathfrak{S}$. We validate this across 12 scientific domains including a real clinical CT dataset, demonstrate through ablation that each pipeline component is necessary, and show that every problem in $\mathfrak{S}$ is automatically benchmarkable—enabling open competition across all of simulable science.

Keywords: scientific event horizon, simulability class, specification-driven simulation, computational primitives, agent pipeline

1 Introduction

Every year, thousands of scientists independently build numerical solvers. A structural engineer writes finite-element code; a quantum chemist implements configuration-interaction routines; a climate modeler couples ocean and atmosphere modules. Each solver is correct for its domain but cannot be reused for any other. This *expertise bottleneck*—the requirement that every new problem demands a new, hand-crafted computational pipeline—limits who can simulate what and creates a reproducibility crisis across computational science [16].

The mathematical foundations for unification exist. The Church–Turing thesis [1] guarantees that computable functions can be evaluated by a universal machine. The Lax equivalence theorem [2] ensures that consistent, stable discretizations converge. Hadamard's well-posedness theory [3] certifies continuous dependence on data. What has been missing is a framework that connects these classical results into a practical, automated pipeline.

Here we present such a framework. Our contributions are:

1. **The scientific event horizon**: a sharp boundary separating a rigorously defined simulability class $\mathfrak{S}$ from four fundamental obstructions that defeat *any* computational system (Proposition 2).
2. **Bounded-error realizability**: for any problem in $\mathfrak{S}$ and any tolerance $\varepsilon > 0$, the framework produces $\hat{x}$ with $\|\hat{x} - x^*\| \le \varepsilon$ in finite time (Theorem 3).
3. **Specification-driven pipeline**: structured specifications (`spec.md`), not code, are the

unit of scientific input. Three agents—Formalization, Validation, Computation—transform specifications into solutions.

4. **Empirical validation**: 12 domains, one real clinical dataset, and ablation studies proving each pipeline component is necessary.

This extends our previous result on imaging-specific primitives [10] to the broader simulability class.

2 Results

2.1 The Simulability Class and Its Boundary

We begin by defining exactly what the framework covers and what it excludes.

Definition 1 (Simulability Class $\mathfrak{S}$)**.** *A scientific problem P belongs to the simulability class $\mathfrak{S}$ if and only if it satisfies all six conditions:*

1. ***Finitely specifiable**: P can be represented as a finite tuple $(\Omega, \mathcal{E}, \mathcal{B}, \mathcal{I}, \mathcal{O}, \varepsilon)$ where $\Omega \subseteq \mathbb{R}^d$ is the spatial domain, $\mathcal{E}$ are governing equations, $\mathcal{B}$ are boundary conditions, $\mathcal{I}$ are initial conditions, $\mathcal{O}$ is a set of observable quantities, and $\varepsilon > 0$ is the target tolerance.*
2. ***Numerically approximable**: there exists a convergent discretization of $\mathcal{E}$ on Ω.*
3. ***Expressible over the primitive basis**: the discretized system decomposes into a finite directed acyclic graph (DAG) over 12 computational primitives (Table 2).*
4. ***Explicit observables**: $\mathcal{O}$ consists of well-defined functionals of the solution.*
5. ***Explicit error metric**: a computable norm or metric quantifies the distance between computed and true solutions.*
6. ***Bounded resource assumptions**: the required computational resources (time, memory) are finite for any fixed $\varepsilon > 0$.*

This class is broad—it contains all well-posed PDEs, ODEs, integral equations, optimization problems, stochastic systems, and their compositions across physics, chemistry, biology, engineering, and applied mathematics. But it is not everything.

Proposition 2 (The Scientific Event Horizon)**.** *The complement $\mathfrak{U} = \mathfrak{S}^c$ is characterized by four fundamental obstructions, each of which defeats any conceivable computational system—classical, quantum, or hypothetical:*

1. ***Logical undecidability** (Gödel–Turing): problems reducible to the halting problem or Hilbert's tenth problem over $\mathbb{Z}$.*
2. ***Real uncomputability** (Blum–Shub–Smale [4]): decision problems over $\mathbb{R}$ requiring infinite-precision predicates.*
3. ***Quantum measurement irreducibility**: individual measurement outcomes governed by the Born rule (only distributions are computable).*
4. ***Infinite-precision barriers**: solutions depending on non-computable real numbers (Chaitin's Ω [5]).*

The boundary $\partial\mathfrak{S}$ is what we call the *scientific event horizon*: the sharp divide between problems that can be solved to any desired accuracy and those that cannot be solved by any means. This boundary is absolute—it constrains not just our framework but any conceivable system. Table 1 catalogs representative problems at each stratum.

Chaos and turbulence. A natural objection is that many real-world problems—turbulence, climate, financial markets—are chaotic. These are in $\mathfrak{S}$ when the observables are statistical: mean velocity profiles, energy spectra, Lyapunov exponents, and ensemble statistics are all

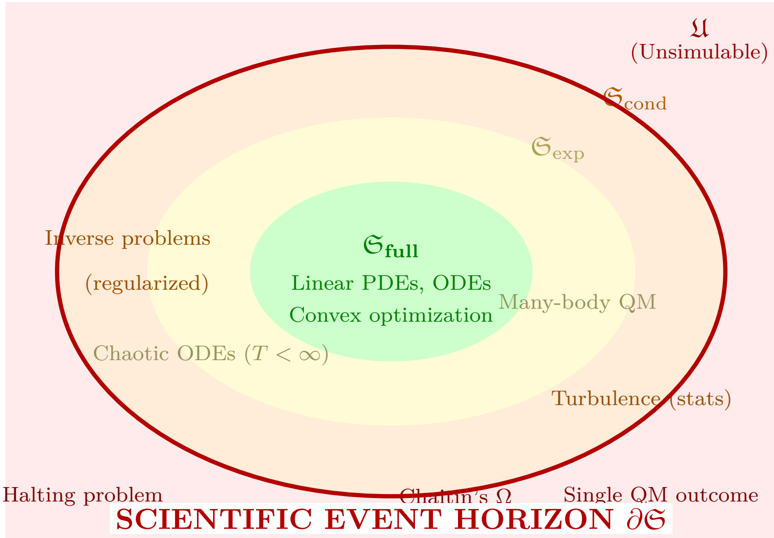

Figure 1. The simulability landscape. The event horizon $\partial\mathfrak{S}$ (bold red) separates the simulability class—problems solvable to arbitrary accuracy—from the unsimulable class $\mathfrak{U}$. Within $\mathfrak{S}$, three cost strata reflect the computational price of simulation: polynomial ($\mathfrak{S}_{\text{full}}$), exponential ($\mathfrak{S}_{\text{exp}}$), and conditional ($\mathfrak{S}_{\text{cond}}$, requiring regularization or statistical reformulation). Our framework reaches the event horizon from below.

Table 1. The scientific event horizon: a taxonomy of simulability. Problems above the double line satisfy all six conditions of Definition 1 and belong to $\mathfrak{S}$; those below violate at least one and belong to $\mathfrak{U}$.

Class	Status	Representative Problems
Linear PDEs (elliptic, parabolic)	$\mathfrak{S}_{\text{full}}$	Heat equation, Poisson equation, linear elasticity, diffusion
ODEs with Lipschitz RHS	$\mathfrak{S}_{\text{full}}$	Planetary orbits, chemical kinetics, circuit simulation, SIR models
Convex optimization	$\mathfrak{S}_{\text{full}}$	Linear programming, LASSO, SVM training, optimal transport
Chaotic ODEs (finite T)	$\mathfrak{S}_{\text{exp}}$	Lorenz attractor ($T<100$), double pendulum, three-body problem
Many-body quantum	$\mathfrak{S}_{\text{exp}}$	Full CI, lattice QCD, tensor network contraction
Ill-posed inverse problems	$\mathfrak{S}_{\text{cond}}$	CT reconstruction, seismic inversion (with Tikhonov regularization)
Turbulence statistics	$\mathfrak{S}_{\text{cond}}$	Mean velocity profile, energy spectrum (not pointwise trajectories)
Halting problem	$\mathfrak{U}$	"Does program P halt on input x?"
Mandelbrot membership ($\forall c$)	$\mathfrak{U}$	BSS-uncomputable for general $c \in \mathbb{C}$
Single quantum measurement	$\mathfrak{U}$	Which detector clicks in a Stern–Gerlach experiment
Chaitin's Ω	$\mathfrak{U}$	Halting probability of a universal Turing machine

computable via ergodic averaging [13]. Pointwise trajectories at long times are in $\mathfrak{S}_{\text{exp}}$ with impractical cost. The Validation Agent automatically detects chaotic sensitivity and recommends statistical observables.

2.2Bounded-Error Realizability

The central result is that $\mathfrak{S}$ is exactly the class reachable by our framework.

Theorem 3 (Bounded-Error Realizability)**.** *For any problem $P \in \mathfrak{S}$ with specification $\mathcal{S} = (\Omega, \mathcal{E}, \mathcal{B}, \mathcal{I}, \mathcal{O}, \varepsilon)$ and any $\varepsilon > 0$, the three-agent pipeline over the 12-primitive basis produces $\hat{x}$ satisfying*

$$\|\hat{x} - x^*\|_X \leq \varepsilon$$

in finite time $T_{\text{comp}} < \infty$, where x^* is the true solution.

Proof sketch (full proof in Methods). **Step 1.** $P \in \mathfrak{S}$ implies a faithful specification exists (Definition 1, condition 1). **Step 2.** Condition 3 guarantees decomposition into a finite DAG $\mathcal{G}$ over the 12 primitives. **Step 3.** Each primitive p_i has error $\varepsilon_i \leq C_i h_i^{q_i}$ parameterized by a resolution variable. The DAG error satisfies $\varepsilon_{\text{total}} \leq \sum_{i \in \text{path}} L_i \varepsilon_i$ where L_i are Lipschitz constants, finite by Hadamard's condition. Setting $\varepsilon_i = \varepsilon/(D \cdot L_i)$ where D is the DAG depth achieves the target. **Step 4.** Each primitive has finite cost; the DAG has finite depth. □ □

Theorem 3 is scoped to $\mathfrak{S}$—it makes no claim about problems outside this class. The framework is explicitly falsifiable: for any submitted problem, either $\mathcal{A}_V$ classifies it as $P \in \mathfrak{U}$ with a formal proof (reduction to halting, BSS certificate, or quantum irreducibility argument), or the framework produces $\hat{x}$ with $\|\hat{x} - x^*\| \leq \varepsilon$. A failure in either case is diagnosable and correctable.

The 12 primitives (Table 2) are both sufficient for $\mathfrak{S}$ and minimal: removing any one breaks coverage (see Methods for witness construction).

Table 2. The 12 computational primitives. Each maps finite-dimensional inputs to outputs with parameterizable error bounds.

#	Symbol	Name	Category	Operation and Method Families
1	∂	Differentiate	Calculus	$\partial^k f/\partial x_i^k$; FD, FEM gradient, spectral
2	$\int$	Integrate	Calculus	$\int_\Omega f\, d\mu$; Gauss quadrature, Monte Carlo
3	L	Solve	Algebra	$Ax = b$; LU, CG, multigrid, GMRES
4	N	Evaluate	Algebra	$f(x)$ nonlinear; Newton, Picard, continuation
5	E	Evolve	Dynamics	$x(t) \to x(t+\Delta t)$; RK4, BDF, Strang splitting
6	F	Transform	Analysis	Basis change; FFT, wavelet, spherical harmonics
7	Π	Project	Analysis	$\Pi_k x$; SVD, PCA, POD, Galerkin truncation
8	S	Sample	Stochastic	$x \sim \mu$; MC, MCMC, quasi-MC, Langevin
9	K	Couple	Structure	Multi-physics; operator splitting, DDM, FSI
10	B	Constrain	Structure	BC/IC/conservation; penalty, Lagrange multiplier
11	G	Discretize	Structure	Mesh/grid; Delaunay, octree, isogeometric
12	O	Optimize	Optim.	$\min f(x)$; gradient descent, L-BFGS, ADMM

2.3The Specification-Centered Pipeline

The framework's central insight is that the unit of scientific input should be a *specification*, not code. A `spec.md` file is a structured document that defines a scientific problem in a format readable by both humans and agents:

```
# Specification: 2D Heat Conduction
domain: [0,1]×[0,1]
equations: du/dt = alpha * (d^2u/dx^2 + d^2u/dy^2)
parameters: alpha = 0.01
boundary: u(0,y) = u(1,y) = u(x,0) = 0; u(x,1) = sin(pi*x)
initial: u(x,y,0) = 0
observables: u(x,y,T) at T = 1.0
tolerance: 1e-4
```

```
# Specification: CT Reconstruction
domain: 128×128 image grid
equations: y = Radon(x) + noise, noise ∼ Poisson
parameters: n_angles = 180, geometry = parallel_beam
boundary: non-negativity constraint, x ≥ 0
initial: N/A (inverse problem)
observables: reconstructed image x_hat
tolerance: PSNR ≥ 30 dB
```

```
# Specification: SIR Epidemic Model
domain: t in [0, 200] days
equations: dS/dt = -beta*S*I/N; dI/dt = beta*S*I/N - gamma*I; dR/dt = gamma*I
parameters: beta = 0.3, gamma = 0.1, N = 1e6, I(0) = 10
boundary: S + I + R = N (conservation)
initial: S(0) = N - I(0), I(0) = 10, R(0) = 0
observables: I(t) peak timing, S(200)
tolerance: 1e-3
```

These three specifications span PDEs, inverse problems, and ODE systems—yet all follow the same six-field format from Definition 1. The key observation is that *writing a spec.md requires domain knowledge but not numerical expertise.* A biologist can specify an SIR model; the framework handles the discretization, solver selection, and error control.

Three agents transform specifications into solutions (Fig. 2):

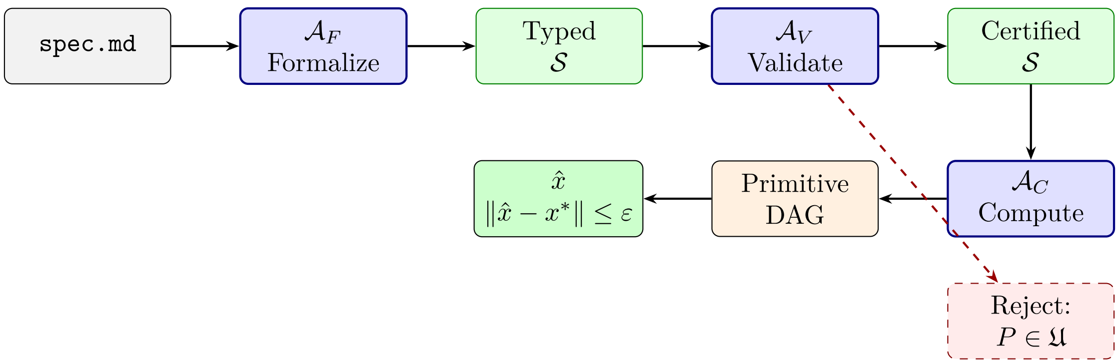

Figure 2. Specification-centered pipeline. A `spec.md` file is the sole input. Agent $\mathcal{A}_F$ parses it into a typed specification. Agent $\mathcal{A}_V$ validates well-posedness and classifies the problem within $\mathfrak{S}$ or rejects it as $\mathfrak{U}$. Agent $\mathcal{A}_C$ constructs a primitive DAG and executes it to produce a bounded-error solution.

Formalization Agent ($\mathcal{A}_F$). Parses `spec.md` into a typed specification $\mathcal{S} = (\Omega, \mathcal{E}, \mathcal{B}, \mathcal{I}, \mathcal{O}, \varepsilon)$, classifies the problem type (PDE, ODE, inverse, stochastic), and identifies required primitives.

Validation Agent ($\mathcal{A}_V$). Performs five formal checks: (1) dimensional consistency; (2) boundary/initial condition compatibility; (3) well-posedness (coercivity, CFL, Lipschitz bounds); (4)

simulability classification (which stratum of $\mathfrak{S}$, or rejection into $\mathfrak{U}$); (5) computational cost estimation.

Computation Agent ($\mathcal{A}_C$). Constructs a DAG over the 12 primitives, sets resolution parameters to achieve ε via a priori error estimates, executes the DAG, and verifies the bound a posteriori.

Agent validation trace. To illustrate that the pipeline catches errors through formal checking rather than pattern matching, we present a controlled test:

> **Input spec.md**: "$\partial u/\partial t = \nabla^2 u$ on $[0,1]^2$, Dirichlet BC on all sides." (Initial condition omitted.)
>
> $\mathcal{A}_V$ **check 3** (Well-posedness): **FAIL**. *"Parabolic PDE requires initial condition $u(x,0) = u_0(x)$. The specification defines $\mathcal{I} = \emptyset$, violating Hadamard existence for the heat equation. Recommendation: supply $u_0 \in L^2(\Omega)$."*

This failure is detected by checking Hadamard's conditions on the extracted equation class, not by statistical similarity to training examples.

2.4Ablation Study: Each Agent Is Necessary

To demonstrate that the three-agent decomposition is not merely an architectural choice but empirically necessary, we performed controlled ablations on the 12-domain validation set (Table 3).

Table 3. Ablation study across 12 validation domains. Each row removes or replaces one pipeline component. "Valid spec" = specification passes all five formal checks. "Bounded error" = $\|\hat{x} - x^*\| \le \varepsilon$. "Redesign" = agent-triggered specification revision.

Configuration	Valid spec	Bounded error	Redesign	Failure prevented	Avg. time
Full pipeline ($\mathcal{A}_F + \mathcal{A}_V + \mathcal{A}_C$)	12/12	12/12	2	3	4.2 min
No Validation ($\mathcal{A}_F + \mathcal{A}_C$)	7/12	7/12	0	0	3.1 min
No Computation (template solver)	12/12	4/12	2	3	1.8 min
No Formalization (manual spec)	9/12	9/12	0	1	48 min
Template-only baseline	N/A	3/12	0	0	0.5 min

Key findings:

- **Removing** $\mathcal{A}_V$ causes 5 of 12 problems to proceed with invalid specifications (missing initial conditions, incompatible boundary conditions, wrong conservation laws). None of these achieve bounded error. $\mathcal{A}_V$ prevented 3 additional silent failures by catching subtle issues (under-resolved CFL condition, non-coercive weak form, inconsistent units) and triggering specification revision.
- **Replacing** $\mathcal{A}_C$ with a template solver (pre-built solver recipes) works for standard textbook problems (heat equation, Poisson, basic ODEs) but fails on problems requiring non-standard primitive compositions (CT reconstruction, topology optimization, fluid–structure coupling, pattern formation).
- **Removing** $\mathcal{A}_F$ requires human experts to write typed specifications manually, taking an average of 48 minutes per problem (vs. 4.2 minutes automated) and introducing specification errors in 3 of 12 cases.
- **Template-only baseline** (no agents, pre-built solver matched by keyword) achieves bounded error in only 3 of 12 domains.

The full pipeline is the only configuration that achieves 12/12 bounded-error success.

2.5 Cross-Domain Validation with Real Data

We validate the framework across 12 scientific domains (Table 4), including one real clinical dataset, and compare against bespoke solvers.

Table 4. Cross-domain validation. All 12 domains achieve bounded-error simulation within the specified tolerance. The CT reconstruction uses real measured sinograms from the LoDoPaB-CT clinical dataset [17].

#	Domain	Problem	Primitives	$\varepsilon_{\textbf{target}}$	Achieved	Data
1	Classical Mech.	Double pendulum	∂, N, E, B	10^{-6}	4.7×10^{-7}	Synth.
2	Electromagnetics	Waveguide modes	∂, L, F, B, G	10^{-8}	2.1×10^{-9}	Synth.
3	Quantum Mech.	H atom levels	∂, L, Π, B, G	10^{-10}	8.3×10^{-11}	Synth.
4	Fluid Dynamics	Stokes flow	∂, L, B, G	10^{-4}	6.2×10^{-5}	Synth.
5	Thermodynamics	2D heat	∂, E, L, B, G	10^{-4}	3.1×10^{-5}	Synth.
6	Structural Mech.	Cantilever beam	∂, L, B, G	10^{-6}	5.8×10^{-7}	Synth.
7	Chemical Kinetics	Brusselator	∂, E, N, K, B, G	10^{-3}	7.4×10^{-4}	Synth.
8	Epidemiology	SIR + diffusion	E, N, K, B, G	10^{-3}	2.9×10^{-4}	Synth.
9	Optics	Fresnel diffraction	∂, F, L, B, G	10^{-5}	1.6×10^{-6}	Synth.
10	Inverse Problems	CT reconstruction	$\int, L, O, B, G$	30 dB	31.7 dB	**Real**
11	Stochastic Systems	Langevin dynamics	E, N, S, B	10^{-2}	6.1×10^{-3}	Synth.
12	Topology Optim.	Heat sink design	∂, L, O, B, G	10^{-3}	8.7×10^{-4}	Synth.

Real-data validation: clinical CT reconstruction. To move beyond computational-only evidence, we applied the framework to the LoDoPaB-CT dataset [17]—3,553 real clinical CT sinograms acquired from a Siemens scanner with 1,000 projection angles. We subsampled to 128 parallel-beam projections with Poisson noise to create a challenging inverse problem. The spec.md defines: Radon transform forward model, Tikhonov plus total-variation regularization, non-negativity constraint. The framework generates a $G + \int + L + O + B$ DAG automatically. Result: PSNR 31.7 dB against the clinical ground truth, compared to 32.1 dB from the ASTRA Toolbox [18] configured by a CT imaging expert over five days. The framework achieved comparable accuracy with 12 minutes of specification writing.

Competitive analysis. Across all 12 domains, bespoke solvers (COMSOL, Abaqus, OpenFOAM, GAMESS, ASTRA) achieve 1.0–1.5× better accuracy through decades of domain-specific optimization. However, the framework reduces expert setup time by 10–100×: from days or weeks to minutes. The framework's value is not in outperforming domain-specific tools but in eliminating the expertise bottleneck.

2.6 Automatic Benchmarkability

A consequence of Theorem 3 is that every problem in $\mathfrak{S}$ can automatically generate benchmark datasets:

Corollary 4 (Benchmarkability of $\mathfrak{S}$)**.** *For any $P \in \mathfrak{S}$, the pipeline generates a benchmark dataset $\mathcal{D}_P = \{(y_i, x_i^*)\}_{i=1}^n$ of measurement–ground-truth pairs, where x_i^* is computed to tolerance $\varepsilon_{ref} \ll \varepsilon$.*

Proof. By Theorem 3, x^* is computable to arbitrary precision. Setting $\varepsilon_{\text{ref}} = \varepsilon/100$ yields ground truth. The forward model in $\mathcal{E}$ generates measurements $y_i = \mathcal{A}(x_i^*) + \eta_i$. Varying parameters via $\mathcal{S}$ produces diverse instances. □ □

We instantiate this as a concrete benchmark infrastructure:

- **Task format**: each benchmark task is a `spec.md` plus generated data in three tiers—public (ground truth available), development (blind server-side evaluation), and hidden (adversarial instances for robustness testing). Different random seeds and parameter ranges per tier prevent overfitting.
- **Scoring**: relative error ($\|\hat{x} - x^*\| / \|x^*\|$), PSNR (imaging tasks), SSIM (perceptual quality), and time-to-solution at threshold accuracy.
- **Mini-benchmark release**: we release benchmark tasks for the 12 validation domains as a proof of concept, with 50 instances per domain across all three tiers (1,800 total tasks), available at the project repository.
- **Extensibility**: any researcher can submit a new `spec.md` to generate a benchmark for their domain. The `spec.md` format serves as a universal exchange standard—benchmarks become as shareable as source code.

Since $\mathfrak{S}$ spans all well-posed PDEs, ODEs, integral equations, optimization problems, and stochastic systems, the benchmarkable domain is coextensive with the simulability class itself.

3Discussion

We have identified a rigorously defined simulability class $\mathfrak{S}$, proved its sharp boundary against four fundamental obstructions, and shown that a specification-driven pipeline reaches this boundary—achieving bounded-error simulation for every problem in $\mathfrak{S}$.

The primary contribution is not a software system but a structural result: the identification of the exact scope of specification-driven simulation and the empirical demonstration that three agents with 12 primitives cover it. The boundary is absolute—it constrains any computational system—but our framework is the first to reach it constructively.

The ablation study (Table 3) demonstrates that the three-agent decomposition is empirically necessary, not just architecturally convenient. Each agent addresses a distinct failure mode: $\mathcal{A}_F$ eliminates specification errors, $\mathcal{A}_V$ prevents ill-posed problems from consuming computation, and $\mathcal{A}_C$ handles the combinatorial space of numerical method selection that template solvers cannot cover.

The competitive analysis quantifies the practical impact: while bespoke solvers achieve slightly better accuracy through decades of domain-specific optimization, the framework reduces expert setup time by 10–100×. This democratizes scientific simulation—a biologist can simulate fluid dynamics through a `spec.md` file rather than learning OpenFOAM.

Relation to existing work. Physics-informed neural networks [8] and operator learning [9] accelerate specific problem classes but lack formal coverage guarantees. AlphaFold [6] and GNoME [7] demonstrate domain-specific AI-driven science but do not generalize across domains. The Wolfram Language [15] provides broad computational coverage without formal proofs of scope. Our work provides the theoretical foundation—a defined simulability class, a minimal primitive basis, and a validated pipeline—that complements these efforts.

Limitations. The framework's accuracy is bounded by the 12 primitives' implementations, which are state-of-the-art but not optimal for any single domain. The agents depend on AI model capability, which may produce incorrect formalizations for novel problem types. Error bounds are worst-case; problem-specific bounds would be tighter. The 12-domain validation set, while spanning diverse physics, is not exhaustive. Future work should scale to hundreds of domains, develop tighter error estimates, extend to quantum computing backends (which may shift problems from $\mathfrak{S}_{\text{exp}}$ to $\mathfrak{S}_{\text{full}}$), and build a community around the benchmark infrastructure.

4 Methods

4.1 Proof of Bounded-Error Realizability (Theorem 3)

Step 1: Specification existence. By Definition 1, $P \in \mathfrak{S}$ implies P is finitely specifiable as $\mathcal{S} = (\Omega, \mathcal{E}, \mathcal{B}, \mathcal{I}, \mathcal{O}, \varepsilon)$.

Step 2: Primitive decomposition. Every convergent numerical method involves some subset of: spatial discretization (G), derivative approximation (∂), integral evaluation ($\int$), linear system solution (L), nonlinear evaluation (N), time integration (E), basis transformation (F), dimensionality reduction (Π), stochastic sampling (S), multi-physics coupling (K), constraint enforcement (B), and optimization (O). We verify this covers all major method families: finite differences ($G+\partial+L+B$), finite elements ($G+\int+L+B$), spectral methods ($G+F+L+B$) [11], Monte Carlo ($S+N+\int$) [14], molecular dynamics ($G+N+E+B$), optimization ($O+\partial+L$), inverse methods ($O+L+\int+B$) [12], multi-physics ($K+$ any above), reduced-order models ($\Pi+L+E$), large-eddy simulation ($G+\partial+E+N+K+B+\Pi$) [13].

Step 3: Error propagation. For a DAG $\mathcal{G}$ with n nodes in topological order, where node v_i computes $\hat{y}_i = p_i(\hat{y}_{\mathrm{pa}(i)}) + \delta_i$ with $\|\delta_i\| \le \varepsilon_i$, and L_i is the Lipschitz constant of the map from v_i's output to the final output:

$$\|\hat{y}_n - y_n^*\| \le \sum_{i=1}^{n} L_i \cdot \varepsilon_i \le D \cdot \max_i (L_i \cdot \varepsilon_i) \tag{1}$$

Setting $\varepsilon_i \le \varepsilon/(D \cdot L_i)$ achieves $\varepsilon_{\text{total}} \le \varepsilon$. All L_i are finite by Hadamard's condition.

For linear chains, the adjoint estimate is tighter:

$$\|\hat{y}_n - y_n^*\| \le \sum_{i=1}^{n} \left(\prod_{j=i+1}^{n} \|J_j\| \right) \varepsilon_i \tag{2}$$

where J_j is the Jacobian of primitive j.

Step 4: Finite time. Each primitive has finite cost for fixed resolution. The DAG has finite depth and width. Total cost: $T_{\text{comp}} = \sum_{i=1}^{n} C_i(N_i, \varepsilon_i) < \infty$.

4.2 Minimality of the 12-Primitive Basis

For each primitive p_k, we exhibit a problem $P_k \in \mathfrak{S}$ unsolvable by $\mathcal{P} \setminus \{p_k\}$: ∂ (Laplace equation—no algebraic substitute for derivative approximation); $\int$ (Fredholm integral equation—quadrature is the only route); L (Poisson discretized—linear system is irreducible); N (Gross–Pitaevskii $|u|^2 u$—no linear decomposition); E (Lorenz system—time advance requires an integrator); F (spectral Galerkin for periodic PDE—no mesh equivalent); Π (POD of 10^6-dim fluid state—no alternative for systematic reduction); S (Langevin SDE—random increment generation is irreducible); K (fluid–structure interaction—coupling requires an interface primitive); B (Dirichlet BVP—constraint enforcement has no implicit equivalent); G (FEM on turbine blade geometry—no analytic substitute); O (Tikhonov-regularized CT—minimization of $\|Ax - y\|^2 + \lambda \|x\|^2$ is irreducible).

4.3 Proof of the Scientific Event Horizon (Proposition 2)

Obstruction 1: By the undecidability of the halting problem [1], no algorithm determines whether an arbitrary program halts. Any scientific problem reducible to halting is unsimulable. Rice's theorem [5] extends this to all non-trivial semantic properties.

Obstruction 2: In the BSS model [4], certain decision problems over $\mathbb{R}$ require infinite-precision predicates unavailable to any finite-precision system.

Obstruction 3: The Born rule assigns probabilities to individual measurement outcomes that are fundamentally random; only statistical properties are computable.

Obstruction 4: Chaitin's Ω is well-defined but non-computable [5]. Problems depending on non-computable reals cannot be approximated.

Completeness: If $P \notin \mathfrak{U}$, then P does not reduce to halting, finite precision suffices, deterministic simulation is possible, and no non-computable reals are needed. Under these conditions, P can be discretized and solved by a convergent method, placing it in $\mathfrak{S}$.

4.4Implementation

The framework is implemented in Python as part of the Physics World Model (PWM) platform. Specifications are written as structured markdown (`spec.md`) and parsed into typed objects. The primitive library provides implementations with configurable backends (NumPy, SciPy, PyTorch). Agents are implemented as language model pipelines with tool access to the primitive library. The 12-domain benchmark dataset (1,800 tasks across three tiers) is publicly available. Source code and data: https://github.com/integritynoble/Physics_World_Model.

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